Flutter Mock Test

Instructions

- Time: **3 Hours**
- Attempt all questions
- Use **Flutter + Dart**
- Use **setState / basic state management only**

Problem Statement

Build a **User Account System using Firebase Authentication** in Flutter.

The app should allow users to **register, login, and access a home screen securely**.

Screens Required

1. Signup Screen

Create a UI with:

- TextField: **Email**
- TextField: **Password**
- TextField: **Confirm Password**
- Button: **Signup**
- TextButton: "Already have an account? Login"

Functional Requirements (Signup)

- Validate:
  - Email must be valid format
  - Password ≥ 6 characters
  - Password & Confirm Password must match
- On clicking Signup:
  - Use Firebase Auth:

    ```
    createUserWithEmailAndPassword()
    ```

  - Show loading indicator while processing
  - On success → Navigate to Home Screen
  - On error → Show error message (Snackbar/Text)

2. Login Screen

UI Components:

- Email field
- Password field
- Login button
- Navigate to Signup screen

Functional Requirements (Login)

- Validate inputs (not empty)
- On Login:

  ```
  signInWithEmailAndPassword()
  ```

- Show loading indicator
- Handle errors:
  - Wrong password
  - User not found
- On success → Navigate to Home Screen

3. Home Screen

UI Components:

- Text: **Welcome**
- Text: **Logged in user email**
- Button: **Logout**

Functional Requirements (Home)

- Get current user:

  ```
  FirebaseAuth.instance.currentUser
  ```

- Display:
  - Email of logged-in user
- Logout:

  ```
  signOut()
  ```

- After logout → Navigate to Login Screen

Navigation Flow

```
Signup → Home
Login → Home
Home → Logout → Login
```

App Behavior

- If user is already logged in:
  - Skip login → Go directly to Home Screen

(Optional Bonus: use `authStateChanges()`)

Error Handling (Must)

Students must handle:

- Invalid email format
- Weak password
- User not found
- Wrong password

Show errors using:

- Snackbar OR Text widget

Loading Indicator (Must)

- Show **CircularProgressIndicator** when:
  - Signing up
  - Logging in

UI Expectations

- Clean layout
- Proper spacing
- Use Material widgets

Packages Required

- `firebase_core`
- `firebase_auth`